Abstract

This Master's Thesis presents a comprehensive study on quantum error correction (QEC) specifically focused on 2D color codes. Using state-of-the-art simulation tools like Stim and Pymatching, we analyze decoder performance on 4.8.8 lattice geometries. The study concludes with a comparative analysis of the results and explores potential improvements for the decoding process.

QEC DECODERS FOR 2D COLOR CODES

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Quantum Error Correction Decoders for 2D Color Codes: Systematic Simulation and Comparison

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